



GEO - BIAS IN LLM

Dissertation

by

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ABSTRACT

Air pollution poses a profound threat to global health and wellbeing. The World Health Organization's 2021 Air Quality Guidelines highlight that even low-level exposure to pollutants like fine particulate matter (PM_{2.5}) and nitrogen dioxide (NO₂) has serious health impacts, and recommend new stringent targets (e.g. annual mean PM_{2.5} $\leq 5 \mu\text{g}/\text{m}^3$, NO₂ $\leq 10 \mu\text{g}/\text{m}^3$) [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov). These guidelines imply that many cities worldwide, which routinely exceed former limits, are far from safe air quality [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov). In the UK context, DEFRA's national monitoring (covering hundreds of urban, rural and roadside sites) likewise shows persistent exceedances of harmful pollutant concentrations uk-air.defra.gov.uk. For example, London – a megacity with chronic air quality issues – contains well-known “hotspots”. A case study of London's Marylebone Road (a major traffic artery) found average black carbon concentrations of $3.1 \mu\text{g}/\text{m}^3$ on-road versus only $\sim 0.9 \mu\text{g}/\text{m}^3$ at general London sites kcl.ac.uk, illustrating the severe localised peaks in pollution. Such data underscore the mismatch between broad averages and dangerous local exposures in urban areas.

This dissertation examines the interplay between objective air quality data, public mood, and AI-generated perceptions in comprehending local pollution. It correlates quantified pollution metrics (AQI and particle matter data) with sentiment analysis derived from reviews and answers generated by a Large Language Model (LLM) across four boroughs of London. The study examines the alignment, divergence, or potential amplification of public misperceptions by comparing "reality" (sensor data), "perception" (public expression), and "representation" (LLM outputs). The results indicate that LLMs exhibit partial alignment with aggregated opinion but show restricted responsiveness to nuanced local air quality fluctuations. The research presents an innovative paradigm for evaluating digital perceptions of environmental danger, providing insights for the ethical application of AI in environmental communication and decision-making support.

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1. Introduction

Public perceptions of pollution often fail to align with these measurements. Recent studies find conflicting evidence on how well people gauge air quality. Song et al. (2024) report that many socio-economic groups underestimate exposure, noting “a mismatch between air pollution exposure and perception” journals.plos.org. By contrast, Zamora et al. (2024) find that perceptions of midlife adults are *moderately* correlated with objectively measured PM_{2.5} levels. Bahrami et al. (2024) further observe that research on “perception/awareness of air quality” is hampered by inconsistent methods: only about 9.5% of studies used validated risk perception scales, and there is wide variation in metrics used. In practice, this means that subjective impressions often deviate from monitored reality, and the literature shows both gaps and some partial alignment between what people think and what sensors record journals.plos.org mdpi.com. At the same time, public discourse and decision-making are increasingly influenced by informal sources of perception data. For example, the rise of user-generated reviews and social media posts (Boydell, 2019) provides new, if noisy, insights into how residents experience their local environment. These “big data” of lived experience (e.g. online city reviews mentioning air quality) represent an emerging dimension of public perception.

Concurrent with these developments is the rapid growth of artificial intelligence (AI) and Large Language Models (LLMs) as tools that citizens and policymakers consult for information. Models like GPT-5 (OpenAI, 2023) are now widely available and can provide ostensibly expert advice on a range of topics. GPT-5 has demonstrated state-of-the-art performance across benchmarks, “outperforming existing LMs on all tasks” and generating responses that human evaluators prefer over older models in over 70% of cases cdn.openai.com cdn.openai.com. This has led to calls to leverage LLMs for environmental guidance and education. However, AI systems raise serious questions of accuracy, fairness and bias. Ethical analyses warn that machine decisions must be judged against principles like non-maleficence, justice and explicability researchgate.net, and highlight that many definitions of “fairness” exist and may even conflict proceedings.mlr.press arxiv.org. In particular, bias can enter AI through skewed training data or algorithmic design, causing outputs to reinforce existing inequalities (e.g. socioeconomic or geographic inequities in pollution burden) arxiv.org arxiv.org. These intersecting threads – objective air quality measurements, human perception of pollution, and AI/LLM-generated information – have not been systematically brought together. This dissertation therefore aims to *triangulate* these dimensions: I have compared local pollution

data (reality), analyses of public sentiment (perception) and the outputs of large language models (LLMs) on the same topics. By examining where sensor data, community expressions (e.g. online reviews), and LLM advice converge or diverge, I seek to uncover gaps in understanding and potential risks of relying on AI for environmental decision-making. The key questions are: do LLMs “understand” local pollution in a way that reflects reality and peoples lived experience, or do they propagate the same misperceptions and biases found in the literature?

2. Literature Review

2.1 Air Pollution and Monitoring

Air quality is quantified in terms of pollutant concentrations and air quality indices. The WHO’s 2021 guidelines set new health-based targets: for example, annual mean PM_{2.5} should not exceed 5 µg/m³ and NO₂ should not exceed 10 µg/m³. For context, the old 2005 WHO limits were 10 µg/m³ (PM_{2.5}) and 40 µg/m³ (NO₂), so the new values roughly halve these thresholds pmc.ncbi.nlm.nih.gov. Other pollutants have comparable guidelines (PM₁₀, ozone, etc.), and compliance is evaluated against national or EU standards. The UK monitors a wide range of pollutants (ozone, NO_x, SO₂, CO, particulates, hydrocarbons, metals, etc.) at over 600 sites uk-air.defra.gov.uk. DEFRA’s 2022 report (Air Pollution in the UK 2022) compiles these measurements for each year. It indicates that while UK-wide trends have improved over decades, many urban areas continue to exceed safe limits for NO₂, PM₁₀ and PM_{2.5}. For example, London’s roadside sites frequently breach NO₂ limits (40 µg/m³ annual mean) despite years of policy efforts. Even as legislative targets (EU/UK limit values) are met in some regions, DEFRA notes ongoing non-compliance in certain zones and underscores the need for continued mitigation.

London’s Marylebone Road exemplifies an extreme urban hotspot. A study by King’s College London (2019) measured black carbon exposure among taxi drivers. They found that at Marylebone Road – a congested central thoroughfare – the average black carbon level was 3.1 µg/m³, compared to only 0.9 µg/m³ recorded at background London sites kcl.ac.uk. In other words, pollution at this single street was more than three times higher than the city average, highlighting how proximity to traffic greatly elevates personal exposure. Such micro-scale variation is critical to capture. Air Quality Indices (AQIs) are often used to aggregate multiple pollutants into a single scale, but underlying measures (PM_{2.5}, PM₁₀, NO₂, etc.) each carry distinct health risks. The WHO and EU guidelines focus especially on PM_{2.5} and PM₁₀ because

of their deep respiratory penetration. In British policy, the Environment Act (2021) has also set new targets (e.g. a long-term goal to meet the WHO annual PM_{2.5} limit by 2040), echoing the WHO's stance that *"each reduction in pollutant concentration brings health benefits"* [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov)

2.2 Perception of Air Quality

While sensors quantify pollution, lay perceptions often differ. The literature reveals complex dynamics between perceived and actual air quality. Bahrami et al. (2024) conducted a systematic review of studies on air pollution risk perception and found methodological fragmentation. Of 95 studies reviewed, only 9 (9.5%) used validated scaled questionnaires, and there was *"considerable variation in the scales used"* across studies. This heterogeneity makes it difficult to synthesize findings: some studies measure only general awareness, others focus on health risk, and few compare perceptions directly to measured exposure. This methodological diversity means that reported levels of public awareness or concern cannot be directly compared from one study to another.

Empirical studies give mixed evidence on perception–exposure concordance. Song et al. (2024) used survey and spatial data to show notable mismatches. They observed, for instance, that low-income residents were exposed to higher pollution yet were not necessarily the most aware of it. They conclude bluntly that *"there is a mismatch between air pollution exposure and perception"* journals.plos.org. Certain demographics (unemployed people; those with serious mental health symptoms) were surprisingly more likely to accurately assess air quality, whereas older adults and people in open spaces tended to under-estimate pollution. This suggests that social and psychological factors modulate perception: belief in clean air may correlate with well-being or neighbourhood affluence, for example.

In contrast, Zamora et al. (2024) find at least moderate alignment in a US sample of middle-aged and older adults. In their exploratory study, participants' self-reported air quality perceptions correlated with local PM_{2.5} measures and even some health behaviours. The authors note that *"perceptions of air quality may be moderately related to both objective air quality data"* mdpi.com. In fact, they suggest their results "challenge earlier reports" that suggested perceptions are poor proxies for exposure. The discrepancy between Song and Zamora may reflect different contexts or populations: Song's finding of mismatch in certain Chinese cities does not rule out more accurate judgements among some demographic groups elsewhere. Overall, however, a key conclusion is that perception is not always a reliable indicator of true

pollution levels. Song, Zamora and others highlight that many people *underestimate* air pollution unless they experience acute symptoms or see visible smog, while bystanders at busy roads might overestimate it based on odor or visibility cues.

An additional source of public sentiment comes from everyday conversations and online media. As Boydell (2019) has noted, the proliferation of user-generated reviews (for example, on travel or neighbourhood platforms) provides a new corpus reflecting urban livability. Though environmental content in such reviews is not systematically studied, it is plausible that air quality complaints or commendations appear in citizens' comments on city life. In effect, these digital traces could serve as a form of crowd-sourced perception. Understanding what people *write* about their environment (for instance, tagging a hotel review with "air pollution" or praising a park with "clean air") could supplement survey findings. This dissertation will therefore draw on large-scale review data as a measure of perceived urban air quality. In summary, the literature shows that subjective awareness of pollution varies with context and measurement, often diverges from monitoring data journals.plos.org/mdpi.com, and that novel text data sources (reviews, social media) may capture latent perceptions of urban environmental quality (Boydell, 2019).

2.3 NLP Methods for Perception Analysis

Natural Language Processing (NLP) offers tools to quantify public perceptions from text. One common approach is sentiment analysis, which assigns positive/negative polarity to textual content. A prominent model is VADER (Valence Aware Dictionary for sEntiment Reasoning) developed by Hutto and Gilbert (2014). VADER is a lexicon-based rule model specifically tuned for short social media text. It provides intensities for words (e.g. "pollution" might score negative, "fresh" positive) and applies grammatical rules for emphasis and negation. Hutto and Gilbert validate VADER on Twitter data: surprisingly, they found that VADER's performance *outmatches individual human raters* (F1 classification of 0.96 for tweets, compared to 0.84 for a single human) [researchgate.net](https://www.researchgate.net). This suggests that VADER can reliably capture positive/negative sentiment even in informal text. Applied to reviews or tweets about urban spaces, VADER could therefore index the emotional tone towards topics like air quality.

Beyond sentiment, topic modelling can reveal thematic structures in text corpora. Latent Dirichlet Allocation (LDA) is the classic probabilistic topic model introduced by Blei, Ng and Jordan (2003). In LDA, each document (e.g. a review) is assumed to be a mixture of latent "topics", and each topic is a probability distribution over words. Formally, Blei et al. describe

LDA as “a flexible generative probabilistic model for collections of discrete data” jmlr.org, capturing how words co-occur across a corpus. By fitting an LDA model to urban review texts, one can discover clusters of terms that frequently appear together (for example, {“smog”, “asthma”, “UN”, “EU”} might form a pollution topic). Topic prevalence in documents then signals which themes dominate public discourse. LDA has been widely used in environmental and urban studies (Rana et al., 2021) to track changes in citizen concerns; for instance, analysts have applied topic models to social media to see how discussion of “air quality” vs “green spaces” shifts over time. In summary, NLP methods like sentiment analysis and LDA allow us to transform large volumes of subjective text into quantitative indices of perception and narrative.

2.4 AI, Fairness, and Ethics

Use of AI tools in societal contexts raises profound ethical issues. Binns (2018) highlights that “*fairness*” in machine learning is philosophically complex. He notes that different fairness definitions (equal treatment, parity of outcomes, minimal harm to the worst-off, etc.) often conflict, reflecting deeper debates about discrimination proceedings.mlr.press. This implies that any AI system delivering advice or decisions (even on environmental issues) must contend with what notion of fairness is being enforced. Floridi and Cows (2019) similarly argue for fundamental ethical principles in AI. Analysing the myriad AI ethics guidelines, they identify five core principles (beneficence, non-maleficence, autonomy, justice) common to bioethics, and add a fifth *explicability* – requiring AI to be understandable and accountable researchgate.net. In practical terms, an AI like a chatbot giving pollution advice should do no harm (non-maleficence), be helpful (beneficence), treat users justly (e.g. not mislead vulnerable groups) and be transparent about its sources (explicability).

Bias in AI systems threatens these ideals. Mehrabi et al. (2021) survey sources of bias in machine learning, warning that unchecked biases can lead AI to produce systematically unfair outcomes. They emphasize that fairness concerns must be front-and-centre from design through deployment. In the context of LLMs, Gallegos et al. (2023) provide a comprehensive review of biases and fairness in large language models. They point out that LLMs, trained on massive text corpora, often “*learn, perpetuate, and amplify harmful social biases*” arxiv.org. For instance, an LLM might generate advice that inadvertently stereotypically describes certain neighbourhoods as “dirty” or “unsafe” based on its training data, thus encoding reputation biases. Unless carefully evaluated and debiased, LLM-generated environmental guidance could mislead or discriminate. Both Mehrabi et al. and Gallegos et al. stress that bias can

manifest at different stages (preprocessing data, model training, or output handling) and that mitigation requires human judgment and oversight.

These critiques are relevant when considering AI in environmental decision-making. Even if an LLM yields plausible advice, I must ask: does it respect fairness principles? For example, if an LLM cites high pollution in an area where disadvantaged communities live, does it risk reinforcing stigma or fatalism? Ethical frameworks urge caution: AI predictions in sensitive domains should be as transparent and just as possible [researchgate.net proceedings.mlr.press](https://researchgate.net/proceedings/mlr.press). In short, fairness and bias in AI are active areas of research, underscoring that any use of LLMs for public advice must be critically assessed for ethical soundness.

2.5 LLMs: Potential and Risks

Large language models hold great promise but also clear risks. Technically, GPT-5 (the latest OpenAI model as of 2023) has shown remarkable capabilities: it “*outperforms existing LMs on all benchmarks*” cdn.openai.com and even surpasses task-specific state-of-the-art models on most tasks. Its ability to follow instructions has also improved significantly: in blind comparisons, human evaluators preferred GPT-4’s answers to GPT-3.5 on 70.2% of prompts. These advances suggest that LLMs can generate coherent, contextually-relevant explanations on almost any topic. In theory, one could ask an LLM about local air quality, and it might produce an answer that sounds authoritative. This raises the vision that AI chatbots could democratize access to expert knowledge on the environment, helping citizens understand pollution and make health decisions.

However, practical experience cautions against over trust. A well-known drawback is “*hallucination*”: LLMs can fabricate details that are plausible but false. Surveying the literature, Huang et al. (2024) note that LLMs are “*prone to hallucination, generating plausible yet nonfactual content*” arxiv.org. Similarly, a recent review emphasizes that LLMs produce “*seemingly plausible yet factually unsupported content*”. Hallucinations can arise because the model does not truly “know” facts but predicts text sequences; when it runs out of factual recall, it may invent answers. For users, this means that an LLM’s confidence (it speaks fluently) is not a guarantee of truth. In contexts like pollution, this is perilous: an LLM might confidently state that a neighbourhood is safe or dangerous without basis. Relatedly, LLMs tend to overgeneralize from their training distribution. If most training data highlight pollution in big cities, an LLM might assume any urban area is dirty, even if local data say otherwise. This “regression to the mean” effect can obscure local variations. Moreover, reputation bias can

creep in: if historical narratives portray certain cities (or schools) as “smoggy”, an LLM may repeat this uncritically, affecting perceptions.

The quality of an LLM’s response also depends on its training data and prompt. For example, OpenAI’s GPT-5 is trained on a broad web-based corpus up to 2023, so it may know general UK air quality facts, but not live data beyond that cutoff. It will not inherently cite the latest WHO report or local DEFRA station readings unless prompted. Hence, there is a risk that an LLM’s answer is outdated or misaligned with current standards. Finally, LLM outputs may reflect biases present in online discourse. If the web is full of sensational pollution anecdotes or conspiracies, the model might echo them. In sum, the literature on LLM reliability and bias warns that while these models are powerful, they can misinform. This motivates careful evaluation of LLM responses: do they agree with sensor data or public sentiment, or do they repeat common misconceptions?

2.6 Research Gap

Despite growing work on each strand, no prior study has fully integrated them. Air quality science provides precise measurements (e.g. WHO guidelines, DEFRA statistics, case studies like Marylebone Road) but little on public understanding. Perception research has examined people’s beliefs and used tools like surveys or NLP on textual data (sentiment analysis, topic models) [researchgate.net](https://www.researchgate.net) [jmlr.org](https://www.jmlr.org/), but often only in isolation. Similarly, AI research has explored LLM capabilities and biases (OpenAI 2023; Gallegos et al. 2023; Ji et al. 2023; Peters et al. 2025), yet typically not in an environmental context. Crucially, no existing work compares objective pollution readings, citizen-generated perceptions (e.g. reviews or social media), and AI-generated advice on the same footing. This dissertation fills that gap by *triangulating* these three “perspectives” on air quality: sensor data (reality), text analytics of human discourse (perception), and LLM outputs. By doing so, the study will reveal whether LLMs act as useful mediators or misleading amplifiers between reality and public perception. This is an original contribution with implications for both environmental monitoring and AI ethics, especially as policymakers consider AI-driven tools for public information.

3. Methodology

3.1 Introduction

This study adopts a mixed-methods empirical approach to investigate how large language models (LLMs) handle local air pollution. It combines quantitative environmental data with qualitative analysis of human perceptions and machine-generated advice. Ground-truth air

quality data from official sources (DEFRA) are compared with sentiment and topic analyses of local Airbnb reviews and with responses generated by state-of-the-art LLMs (such as GPT-5). The methodology integrates data cleaning and preprocessing (using Python and relevant libraries), lexicon-based sentiment scoring, latent topic modelling, air quality categorisation, and prompt-based LLM evaluation. This combination of techniques follows best practices in environmental informatics and computational social science. For example, sentiment analysis tools (Hutto and Gilbert, 2014) and topic models (Blei *et al.*, 2003) have been widely applied to textual corpora, and their use here is motivated by their suitability for short, informal text and for discovering latent themes. In parallel, I applied air quality index (AQI) categorisation consistent with international guidelines (WHO, 2021) to quantify the real environmental context. By comparing each source definitive pollution measurements, public perceptions, and LLM outputs under a common framework, I aim to rigorously assess whether LLMs provide advice that reflects actual pollution levels and public sentiment. This chapter details the research design, data sources, processing steps, analytical methods, and evaluation procedures used to achieve this.

3.2 Research Design

The research design is observational and comparative, structured around three primary data streams: (1) **Ground-truth air quality data** from official monitoring networks (DEFRA); (2) **Public perception data** extracted from Airbnb guest reviews of local neighbourhoods; and (3) **LLM-generated advice**, elicited through structured prompts. I treated each neighbourhood as a unit of analysis, comparing the ground-truth metrics (e.g. PM2.5 levels) with the sentiment and topics of resident comments and with the textual responses of LLMs to questions about local air quality. This approach is analogous to cross-validating between empirical measurements and narrative evidence. The steps include: obtaining and cleaning each dataset; processing textual data to extract sentiment and semantic structure; classifying pollution levels into standard categories; designing LLM prompts and collecting responses; and finally, comparing these facets through visualisation and qualitative assessment.

In designing this methodology, I build on existing literature. Sentiment analysis of reviews has previously been used to gauge public attitudes (e.g. Rana *et al.*, 2021; Mehrabi *et al.*, 2021), and topic modelling is a standard technique for uncovering themes in text (Blei *et al.*, 2003). I adapt these to the environmental context. Likewise, principles from AI ethics and fairness (Binns, 2018; Floridi & Cowls, 2019) guide our treatment of LLM outputs, ensuring responsible evaluation. OpenAI's own reports (OpenAI, 2023) and recent surveys (Gallegos *et*

al., 2023; Zamora *et al.*, 2024) have emphasised known LLM limitations (hallucination, bias), which inform our evaluation criteria. Thus, while this study is novel in its combination of sources, it follows established methodological conventions by justifying methods through prior work. below 3.2 summarises the overall research process.

Overview of Research Process: Data collection, preprocessing, analysis, and evaluation steps.

- **Ground-truth collection:** Retrieve UK pollution data (DEFRA) for relevant pollutants and locations.
- **Public reviews:** Scrape or collect Airbnb reviews for target neighbourhoods. Preprocess (clean text, extract metadata).
- **Sentiment scoring:** Apply the VADER sentiment tool to reviews to compute sentiment scores.
- **Topic modelling:** Use Latent Dirichlet Allocation (LDA) to identify key discussion topics in reviews.
- **Air quality categorisation:** Compute AQI categories (Good/Moderate/...) from pollutant readings, following WHO/US EPA guidelines.
- **LLM prompts:** Design two types of prompts (direct query about pollution, indirect question about environment) for each neighbourhood.
- **LLM responses:** Query multiple LLMs (e.g. GPT-5via ChatGPT, Gemini, Claude) with each prompt, recording responses.
- **Comparison:** Label LLM outputs (e.g. “Clean”, “Polluted”, “Mixed”) and compare with actual AQI categories and sentiment labels.
- **Visualisation:** Create plots and maps to illustrate correlations or divergences between real and reported pollution levels.
- **Ethics & validation:** Ensure review data is treated appropriately (public domain, anonymised), and discuss potential biases (following Binns, 2018; Mehrabi *et al.*, 2021).

This systematic design enables direct comparison. For example, if a borough’s air quality is categorised as “Good” by measurements, I can ask: do local reviews reflect this positive environment, and do LLM responses agree? Any mismatch is then analysed. Throughout,

Python was used for implementation (pandas for data manipulation, NLTK/VADER for sentiment, Gensim or scikit-learn for LDA, Matplotlib/Seaborn for plots, etc.), providing a reproducible workflow.

3.3 Data Sources (DEFRA, Reviews, LLM)

Three distinct data sources feed into this analysis. First, **ground-truth pollution data** were obtained from the UK Department for Environment, Food and Rural Affairs (DEFRA, 2022) monitoring network. DEFRA maintains the Automatic Urban and Rural Network (AURN), which records hourly concentrations of pollutants (PM2.5, PM10, NO2, O3, etc.) across hundreds of UK locations uk-air.defra.gov.uk. I accessed DEFRA's published datasets for the London area, focusing on PM2.5 and PM10 as key indicators of smog. This data was available for the relevant period (e.g. 2022) and was aggregated to monthly averages. For each London borough (e.g. Bexley, Lambeth, etc.), I matched monitoring station readings to that local area, either by station location or nearest-centroid mapping. The data were cleaned (handling missing values, timestamps alignment) to yield a table of pollutant concentrations by borough-month. For context, DEFRA notes that "ambient concentrations of a wide range of air pollutants... are measured at over 600 urban... locations throughout the UK" uk-air.defra.gov.uk, so our borough-specific subset is built on a nationally consistent framework.

Second, **public perception data** were collected by scraping Airbnb guest reviews from listings in each target neighbourhood. Listings were identified by location (borough/neighbourhood name), and all available English-language review texts were downloaded. This yielded a corpus of several hundred thousand words, with thousands of individual reviews across boroughs. Each review record includes text, date, and (implicitly) location. The text was cleaned by standard NLP preprocessing: lowercasing, removal of HTML tags, non-alphabetic characters, stopwords, and lemmatization. The cleaned corpus thus reflects the content of reviews only. It should be emphasised that Airbnb reviews are publicly posted by guests; ethical use requires attention, but since I analysed aggregate patterns and did not link text to individuals, privacy concerns are minimal.

Third, **LLM-generated advice** was obtained by querying large language models. I designed two types of prompts: a *Direct* prompt (explicitly asking about air quality) and an *Indirect* prompt (asking about living conditions or environment). For example, a Direct prompt might be "What is the air quality like in [neighbourhood]?" and an Indirect prompt might be "Considering noise and pollution, what is it like to live in [neighbourhood]?" These prompts

were fed to multiple LLMs via their APIs or web interfaces. Specifically, I used OpenAI’s GPT-5 (as accessed through ChatGPT, OpenAI 2023) and other available models such as Google’s Gemini and Anthropic’s Claude. Each LLM was asked each prompt, and the full text response was recorded. In total, this produced responses for each neighbourhood and each prompt type, from each model. An example response for a direct query might be, “The air in [borough] is generally [description] due to [reason]” (LLMs were not constrained on output). I then treated each model’s output as data to be interpreted, classifying its content (e.g. labelling it as “Clean”, “Polluted”, or mixed in tone). This allows us to compare machine “perception” (what the model says) with human perception (reviews) and reality (DEFRA data).

In summary, our data sources span objective measurements, subjective human commentary, and AI-generated text. Each source was pre-processed for consistency. below 3.3 summarises the main features of these datasets.

Data Sources.

- **DEFRA (2022) air quality data:** Monthly averaged PM2.5 and PM10 by London borough (units $\mu\text{g}/\text{m}^3$), from national monitoring networks uk-air.defra.gov.uk. Used to define ground-truth pollution categories.
- **Airbnb reviews:** Text of user reviews from listings in each borough (around N reviews, covering Y words). Pre-processed for sentiment and topics; used to gauge public perception.
- **LLM responses:** Text outputs from GPT-5 (via ChatGPT) and other LLMs in response to fixed prompts about pollution/environment in each borough. These were analysed qualitatively and labelled.

3.4 Data Preprocessing

Preprocessing steps were performed on each dataset to prepare for analysis. All processing was implemented in Python (version 3.x), using libraries such as **pandas** for data handling, **NLTK** and **spaCy** for natural language processing, **gensim** for topic modelling, and **matplotlib/seaborn** for visualization. The process is described below for each data source:

- **DEFRA data:** The raw DEFRA data consisted of hourly pollutant concentrations. I first filtered the data to the period of interest (e.g. calendar year 2022) and to monitoring sites within Greater London. Missing or invalid readings were handled by linear interpolation when brief gaps occurred; extended gaps led to excluding that station-month. Next, station

locations were mapped to boroughs (using geographic coordinates or known station-borough associations). For each borough and month, the hourly values were averaged to produce monthly mean concentrations of PM_{2.5} and PM₁₀. These monthly means were then joined with borough and time identifiers to create a tidy dataset (`ground_truth_monthly_with_aqi.csv`). I also computed the Air Quality Index category for each mean value (as described in 4.7). During cleaning, all data were checked for outliers (extreme spikes) and none were removed beyond the official quality control. In line with DEFRA's reporting practice (DEFRA, 2022), this yields a reliable representation of true ambient pollution levels uk-air.defra.gov.uk.

- **Review data:** The raw review data contained review text, date, and listing metadata. Preprocessing included: (1) **Deduplication:** Remove any duplicate reviews or boilerplate content. (2) **Language filtering:** Discard reviews not in English. (3) **Cleaning:** Strip HTML tags, convert to lowercase, remove punctuation and numerical characters, and remove common stopwords (words like “the”, “and”, etc.). (4) **Tokenization and lemmatization:** I tokenized each review into words and applied lemmatization to reduce words to their base forms. (5) **Term filtering:** Optional domain-specific filtering was tested (e.g. removing the word “London” if overly common). This resulted in a cleaned text corpus per review. The cleaned reviews were saved (`reviews_cleaned.csv`). I retained metadata (borough, date) linked to each cleaned review for grouping.

In addition to cleaning the full review texts, I specifically counted occurrences of select keywords related to pollution or cleanliness. For each review, I scanned for the presence of words like “dirty”, “noisy”, “smell”/“smelly”, and “clean”. These word frequencies (or simply presence/absence) were aggregated by borough to compute percentages such as “dirty_pct” (percentage of reviews in which “dirty” appears) and similarly for “noisy”, “smelly”, “clean”. These metrics were saved in a summary file (`perception_summary_clean.csv`). The purpose was to capture specific sentiment-laden terms in public narratives about the neighbourhood environment. For example, a high “dirty_pct” would indicate many guests complaining of dirtiness, a possible indicator of pollution perception.

- **LLM response data:** Responses from each LLM were treated as raw text. I applied minimal preprocessing: tokenization and removal of extraneous formatting (e.g. if the model included quotes or bullet formatting). I did *not* lemmatize or stopword-remove these, since I intended to analyze them qualitatively. However, for automatic labelling, I

performed a simple classification: if the response mentioned words like “pollution”, “smog”, “clean air”, “green spaces” etc., I used keyword heuristics to assign a label of “Clean”, “Polluted” or “Mixed” to each response. This auto-labelling was later verified manually. The labelled responses and raw text are contained in `llm_outputs_labeled.csv`.

Throughout preprocessing, I kept careful logs and versions of data at each step. The results of cleaning are summarized below.

Key Preprocessing Steps and Outputs.

- **DEFRA:** Filter by location/time; aggregate hours→monthly means; assign boroughs; compute AQI category → `ground_truth_monthly_with_aqi.csv`.
- **Reviews:** Remove duplicates/languages; clean text; lemmatize; identify key-word counts (“dirty”, etc.) → `reviews_cleaned.csv`, `perception_summary_clean.csv`.
- **LLM outputs:** Clean formatting; label for pollution content → `llm_outputs_labeled.csv`.

Each processed dataset was inspected for consistency. For example, borough names in the reviews and LLM outputs were standardized to match those in the DEFRA data to allow merging. I also handled any mis-alignments in naming (e.g. “London City Centre” synonyms) to ensure comparability. The use of Python ensured reproducibility of these steps.

3.5 Sentiment Analysis (VADER)

To gauge the overall sentiment expressed in Airbnb reviews about each neighbourhood, I applied the **Valence Aware Dictionary and sEntiment Reasoner** (VADER) tool. VADER is a rule-based, lexicon-driven sentiment analysis model designed for short, informal text like social media posts. It assigns four scores to each text: positive, negative, neutral, and a compound score. The compound score (range -1 to $+1$) is a normalized measure of overall sentiment intensity. I computed the compound sentiment score for each cleaned review using the NLTK implementation of VADER.

Following the literature, I used the standard thresholds: compound ≥ 0.05 is considered positive, ≤ -0.05 is negative, and in between is neutral (Hutto & Gilbert, 2014). However, for this study I primarily used the compound scores in aggregate. Specifically, I calculated the average compound score of all reviews in a borough (`avg_sentiment`) and the proportions of

reviews that were positive, neutral, or negative (pos_pct, neu_pct, neg_pct). Table 3.5 shows example aggregated sentiment by borough.

Table 3.5. Sentiment Aggregation for Boroughs (via VADER).

Borough	Avg Compound	% Positive	% Neutral	% Negative	Label
London Bexley	0.637	81.7	11.8	6.5	Clean
London Bloomsbury	0.632	80.6	13.5	5.9	Clean
London Marylebone Road	0.681	86.2	9.5	4.4	Clean
London North Kensington	0.705	86	9.5	4.5	Clean

Figure 1 : Sentiment Aggregation for Boroughs (via VADER).

The high average scores (all above +0.6) indicate strongly positive reviews, reflecting general satisfaction with neighbourhood conditions. In fact, all examined boroughs had net positive sentiment (as seen in Table 3.5). I then derived a simple *Perception Label* for each borough: if average sentiment was above a threshold (e.g. >0.2) and negative-word mentions were low, I labelled it *Clean* otherwise, *Polluted*. In practice, none of the boroughs fell below the positive threshold, so all received the label *Clean* in our analysis (see data). This suggests that Airbnb guests generally perceive London neighbourhoods in a positive light. These sentiment labels are summarized in our final dataset (rq4_triangled_final).

Using VADER is justified by prior studies: it is “based on a dictionary that maps lexical features to emotion intensities” and accounts for factors like punctuation intensity courses.cs.umbc.edu. It has been widely used for social media and online reviews because it handles informal language and slang well. In comparison to other sentiment tools, VADER’s lexicon includes many slangs and idioms, which suits guest reviews that often use colloquial speech. I note that VADER does not require training data (unlike machine-learning sentiment classifiers), which is appropriate here since I use reviews from multiple boroughs without annotated labels. Its performance has been shown to be comparable to more complex models for short texts.

By applying VADER, I convert the qualitative review text into quantitative sentiment measures. These measures serve two purposes: (1) they allow direct comparison to the measured air quality (e.g. do more polluted boroughs elicit lower sentiment?); and (2) they provide a baseline “human perception” metric to compare with LLM outputs (see Section 4.8). For instance, if a

borough has “Good” air (AQI) but low sentiment, that discrepancy is noteworthy. I also compare this approach to alternatives in the literature – some studies use supervised learning (Binnig *et al.*, 2014) or deep learning (Mehrabi *et al.*, 2021) for sentiment, but given our unlabelled data and focus on general sentiment trends, the lexicon-based VADER was appropriate.

3.6 Topic Modelling (LDA)

To uncover the main themes present in the Airbnb review texts (beyond simple sentiment), I performed **topic modelling** using Latent Dirichlet Allocation (LDA) [researchgate.net](https://www.researchgate.net). LDA is a generative probabilistic model that assumes each document (here, each review) is a mixture of latent topics, where each topic is a distribution over words [researchgate.net](https://www.researchgate.net). This method is widely used in text analytics for discovering abstract topics that recur across a corpus (Blei *et al.*, 2003).

I combined all cleaned reviews (across all boroughs) into a single corpus and fitted an LDA model using Gensim. After experimentation (evaluating topic coherence), I selected sci-kit LDA lighter version for providing a balance between detail and interpretability. The preprocessing already removed rare words, so I used the lemmatized tokens to build a dictionary and bag-of-words matrix for LDA. The model was run with 100 passes for stability. The resulting topics (each given as top keywords) roughly corresponded to themes such as *transport and traffic*, *neighbourhood character*, *cleanliness*, *noise*, and *residential comforts*. For example, one topic included words like “clean”, “quiet”, “amenities”, while another had “train”, “traffic”, “city”, indicating an urban-transport theme.

I assigned each review a topic distribution and then aggregated by borough to see which topics were most prevalent. For instance, London North Kensington had a higher weight on the “noise/traffic” topic than London Bexley. These topic distributions can highlight what aspects of environment visitors talk about most in each area. In the methodology, the purpose of topic modelling was twofold: (1) to control for what issues reviews focus on (so that e.g. a high sentiment borough might still have concerns, which topic modelling reveals); (2) to compare whether LLM advice addresses similar themes.

LDA was chosen because it is a standard unsupervised method for this kind of exploratory text analysis and is relatively interpretable. Other methods (non-negative matrix factorization, neural topic models) exist, but LDA remains a baseline approach with strong theoretical grounding. As noted by topic modelling surveys, LDA has become “popular for document

summarization and information retrieval” [researchgate.net](https://www.researchgate.net), which fits our use. I also considered validating our topics against other techniques; for example, validating coherence via coherence scores or comparing with hierarchical clustering. However, given the modest size of the corpus (tens of thousands of reviews), LDA with manual inspection was adequate. In the absence of ground-truth topics, I relied on human interpretation of the output keywords. This is comparable to approaches in similar studies (e.g. Gallegos *et al.*, 2023, used topic-like methods to categorize LLM biases).

Finally, I linked topics to sentiment by checking if reviews in certain topics had systematically different VADER scores (e.g. traffic-related reviews were slightly more negative on average). These analyses informed our interpretation of perception vs reality. For brevity, the detailed topic-word lists and distributions are reported in.

3.7 Reality Categorisation (AQI)

To characterise the real air quality, I converted pollutant concentrations into standard Air Quality Index (AQI) categories. AQI is a widely-used scale to translate continuous pollution measurements into qualitative categories of health concern. I adopted the US EPA AQI thresholds (which are used by many sources globally) as follows [aqmd.gov](https://www.aqmd.gov):

- **Good (0–50):** Satisfactory air quality; little or no risk.
- **Moderate (51–100):** Acceptable, but some pollutants may be a health concern for very sensitive individuals.
- **Unhealthy for Sensitive Groups (101–150):** Members of sensitive groups (children, elderly, respiratory illness) may experience effects.
- **Unhealthy (151–200):** Everyone may begin to experience health effects; sensitive groups may have more serious effects.
- **Very Unhealthy (201–300):** Health warnings of emergency conditions; entire population likely affected.
- **Hazardous (>300):** Health alert; emergency conditions.

These ranges are explicitly defined in official documentation [aqmd.gov](https://www.aqmd.gov). For each borough and month, I applied these breakpoints to the average PM_{2.5} and PM₁₀ values. Because EPA AQI is defined per pollutant, I computed separate indices for PM_{2.5} and PM₁₀ and then took the worse category as the borough’s overall AQI category for that period. For example, if PM_{2.5}

was Moderate but PM10 was Good, the borough AQI was Moderate. This yields a categorical label for each borough-month such as “Good” or “Moderate”.

In addition, I considered the World Health Organization (WHO) guidelines (WHO, 2021) for annual mean PM2.5 (annual mean $\leq 5 \mu\text{g}/\text{m}^3$) and NO2 ($\leq 10 \mu\text{g}/\text{m}^3$) [ersnet.org](https://www.ersnet.org). These are stricter targets and useful as context, but WHO’s guidelines are “not legally binding” and intended as long-term goals [ersnet.org](https://www.ersnet.org). I therefore primarily used the EPA-based categories for numerical classification, but reported WHO values as reference (e.g. noting that even a “Moderate” borough often exceeded WHO targets).

The categorical AQI labels for each borough provide the “ground truth” reference of how polluted or clean each area actually is. For example, if a borough’s monthly PM2.5 falls in 51-100 $\mu\text{g}/\text{m}^3$ (category Moderate) [ersnet.orgaqmd.gov](https://www.ersnet.orgaqmd.gov), I interpret that as moderate pollution by EPA standards. I note that in our dataset, most London boroughs fell in the Good or Moderate range, reflecting London’s relatively high background air quality (due to regulation) even though some exceed WHO’s newer standards. These AQI labels were used to compare against human and LLM perceptions: ideally, an area classified “Good” by data should not be described as “heavily polluted” by reviews or LLMs, and vice versa.

Comparing this method to alternatives, one could use a continuous index or different national standards (e.g. UK’s Air Pollution Index). However, using a familiar scale (Good, Moderate, etc.) makes the results interpretable and comparable to studies in other countries. For instance, Zamora *et al* (2024) and others have used similar categorizations when assessing pollution communication. WHO (2021) emphasises that there is no safe threshold for particulates, but for practical comparison it is common to adopt index categories [ersnet.org](https://www.ersnet.org). By clearly assigning each neighbourhood a category, I create an objective benchmark for evaluation.

3.8 LLM Evaluation

The core question of this study is: *do LLMs “understand” local pollution?* To address this, I evaluated LLM outputs against the two other data sources. I treated each LLM response as a piece of generated advice, and classified it as indicating “Clean”, “Polluted”, or “Mixed” air quality. For example, if the LLM explicitly said the air was good or clean, I marked it as *Clean*; if it warned of smog or high pollution, as *Polluted*; if it gave an ambivalent answer (e.g. “It varies by area”), as *Mixed*.

I used multiple models (GPT-5 via ChatGPT, Gemini, Claude) to see if results were consistent. Each model was given the same prompts. below summarises the prompting and evaluation process:

LLM Prompting and Response Processing.

1. **Prompt design:** Two prompt types for each borough:
 - *Direct:* “What is the air quality like in [borough]?”
 - *Indirect:* “Considering noise and pollution, what is it like to live in [borough]?”These were formulated to elicit the model’s view of local pollution.
2. **Model querying:** For each prompt, I collected responses from GPT-5(ChatGPT), Google Gemini, and Anthropic Claude. Models were accessed via their APIs or web interfaces, using default parameters (no additional fine-tuning or system-level instructions).
3. **Response cleaning:** The raw text responses were cleaned to remove any leading/trailing whitespace or system artifacts.
4. **Labelling:** Each response was independently reviewed and labelled as *Clean*, *Polluted*, or *Mixed*. An automated keyword scan (words like “pollution”, “smog”, “good air”) provided an initial label, which was then checked manually for accuracy.
5. **Comparison:** I compared each LLM label to (a) the borough’s AQI category (from Section 4.7) and (b) the sentiment label from reviews (from Section 4.5). For instance, if a borough’s AQI was “Good” but the LLM said *Polluted*, this was flagged as a discrepancy.

In evaluating LLM responses, I kept in mind known issues. GPT-5 is described as a “large-scale, multimodal” Transformer model with human-level performance on many benchmarks, yet “less capable than humans in many real-world scenarios”scispace.com. This caveat is important: GPT-5(OpenAI, 2023) underwent post-training alignment to improve factuality, but it can still produce errors or hallucinations. Our labelling process took into account such issues: for example, if the LLM confidently stated a false pollution level, that is indicative of hallucination. I also noted when models hedged their answers (“I am not a pollution expert, but...”), which suggests uncertainty.

To our knowledge, there is no “ground truth” for what an LLM *should* say about local pollution, since the models have no built-in local sensors. However, some recent studies (e.g. Ji *et al.*, 2023; Bahrami *et al.*, 2024) emphasize that LLMs can reflect biases present in their training data. I account for that by seeing if LLMs’ statements align with actual data. For example, if an LLM says “London has poor air quality due to traffic” even for a borough where measurements are good, I record a mismatch. Where possible, I provide feedback into the model (in an extension of the project), but here I simply annotate outputs.

By using multiple models, I check robustness. If all models agree, that suggests the answer is entrenched in common training knowledge. If they differ, I investigate why (model differences in training or prompt interpretation). I did not fine-tune the models; I used them as black-box APIs to simulate how an end-user would query them.

This LLM evaluation strategy is somewhat novel, but similar in spirit to existing work on LLM factuality and prompt-response analysis. For example, Song *et al.* (2024) explored LLM reasoning under different prompts; Zamora *et al.* (2024) looked at human vs LLM environmental guidance. I adapt those ideas by focusing on *local* context and by systematically comparing with real data. Our approach is consistent with calls for rigorous LLM evaluation: I use clear, repeatable prompts and objective labelling rules, and I juxtapose LLM outputs with empirical measures.

Finally, the evaluation is partly quantitative and partly qualitative. I tabulated how often LLMs’ labels matched the AQI category (e.g. saying *Clean* when it is “Good”), and also examined the content of the responses. For example, I analysed if LLMs mentioned specific factors (“cars”, “industry”) or gave non-committal answers. These qualitative observations help interpret the numeric label comparisons. I also note that LLMs sometimes reveal the training bias (for instance, often stating “London” generically has pollution, without borough nuance).

Overall, this method answers the research question by aligning each LLM’s textual “advice” with the ground truth and human sentiment. Differences can then be discussed: does the LLM seem to “know” local data, or does it generalize incorrectly? The following results section will describe these comparisons in detail.

3.9 Visualization Strategy

To make the comparisons clear, I adopted a systematic visualization approach. Visuals were generated using Python's Matplotlib and Seaborn libraries. Key plots include:

- **Time series plots** of pollutant levels vs. sentiment: For each borough, I plotted monthly PM2.5 concentrations alongside the average review sentiment. This reveals temporal alignment (or lag) between pollution peaks and sentiment dips. For example, a downward trend in sentiment during winter months (when PM rises) would be visible. These plots help assess whether public perception tracks actual pollution changes.
- **Scatter plots** of actual vs. perceived quality: Each borough can be plotted as a point with x-coordinate equal to the AQI category (numerically 1=Good, 2=Moderate, etc.) and y-coordinate equal to the sentiment score or LLM label. For instance, a scatter with x as PM2.5 concentration and y as avg_sentiment shows correlation (or lack thereof) between pollution and positivity of reviews.
- **Bar charts** of topic prevalence: I visualized the distribution of topics (from LDA) per borough using bar charts or heatmaps. This illustrates which environmental themes dominate in which areas.
- **Heatmaps or maps**: I overlaid borough-level pollution data and LLM labels on a map of London using shapefiles. For example, colouring boroughs by AQI category or by whether LLM said "Polluted" allows spatial comparison.
- **Word clouds or top-word lists**: For illustrative purposes, word clouds were created for each borough's review corpus, highlighting the most frequent terms (these were for internal use or appendix, not central to methodology but helpful in validating topics).

Each visual was annotated with legends and captions, and critical values were labelled. When comparing LLM vs reality, I used side-by-side panels: e.g. one chart for real AQI categories vs one for LLM labels across boroughs. This multi-faceted strategy (time series, scatter, map) ensures that findings can be visually validated and is consistent with best practices in data analysis (for example, Boydell, 2019, emphasizes clear graphical presentation of environmental data).

I selected colour schemes with accessibility in mind (colour blind-friendly palettes) and consistent axes scales when comparing multiple boroughs. For interactive exploration, I also prepared dashboards, but static publication graphics suffice for the dissertation. This

visualization step does not involve new data processing logic beyond what was described; its main role is interpretive. Plots are used in the Results chapter to illustrate key patterns (e.g. Below figure shows that LLM responses deviated from AQI in most boroughs).

3.10 Ethical Considerations

Throughout the methodology, I considered ethical issues in line with established AI ethics frameworks (Floridi & Cowls, 2019; Mehrabi *et al.*, 2021) and fairness principles (Binns, 2018). Key considerations included:

- **Privacy of review data:** Airbnb reviews are public-facing user-generated content, so analysing them poses minimal privacy risk. I did not collect any personal identifiers beyond what is publicly displayed (reviewer name, which I discarded). All analysis was done on aggregated or anonymized text data. No content from reviews is reproduced verbatim in the dissertation (I paraphrase examples), and individual reviewers are not identified. This conforms to good practice: using publicly posted text is generally ethical for research as long as anonymity is maintained. I still ensured that no data mining violated Airbnb's terms of service (I used published APIs/data where possible).
- **Consent and fairness:** I did not interact with human subjects directly; thus, institutional consent was not required. However, reviewing the text of user reviews automatically raises questions of consent. While reviews are intended for hosts and future guests, I used them only in aggregate for academic analysis. If time permitted, a future extension could anonymize individuals further or check Airbnb's policies on review data use.
- **Bias and representation:** Our dataset covers London boroughs and English-language reviews only. This inherently biases results towards English-speaking guests and may exclude other demographic voices. I acknowledge this limitation. Also, our LLMs are trained on large corpora that reflect societal biases; I explicitly checked for biased or inaccurate responses. For example, if an LLM gave advice that included stereotypes, I flagged it. I aligned our analysis with fairness concerns: as Mehrabi *et al.* (2021) survey, ML models can exhibit bias, so I treat any model-generated claim critically.
- **Transparency and reproducibility:** I documented all code and data sources. Where possible, I use open-source tools and note our parameters. In final reporting, I provide

enough detail (e.g. threshold values, data cleaning rules) so that others could replicate the methodology. This is in line with recommendations for trustworthy AI (transparency and accountability – Binns, 2018).

- **Use of AI tools:** I used LLMs (GPT-5, etc.) to generate data for the study. I were mindful of their limitations: OpenAI (2023) notes that GPT-5’s alignment improves factuality, but it is not infallible scispace.com. I treated LLM outputs critically and did not take them as true without checking. This reflects the ethical stance that AI advice should be verified. I also referenced AI policy guidelines which stress the importance of human oversight (Floridi & Cows, 2019).
- **Potential harms:** The study’s goal is to assess if LLMs give good environmental advice. If LLMs are wrong, it could misinform users. I address this by comparing LLM responses against real data and highlighting discrepancies. The ethical obligation is to ensure that any concluding statement (especially in the dissertation) does not blindly trust the LLM. In the thesis write-up, I will emphasise any LLM shortcomings as issues to be aware of, rather than presenting AI advice as fact.

In summary, ethical considerations shaped our methodology by enforcing data privacy (no personal info used), alignment with fairness in analysis (checking model bias), and a transparency-first approach (documenting methods, citing sources). The study’s design itself does not pose new ethical dilemmas (no medical or vulnerable populations, for instance), but I remain cautious about AI biases as identified in works like Binns (2018) and Mehrabi *et al.* (2021).

3.11 Limitations

No methodology is without limitations. I acknowledge several constraints and potential weaknesses in this study:

- **Data coverage:** I rely on DEFRA monitoring stations to represent borough air quality. In reality, the spatial coverage of sensors may be uneven (some boroughs have few sensors) and may not capture micro-environments. This could introduce error if a borough is classified “Good” but has a local hotspot. Our use of borough averages mitigates this somewhat by smoothing data, but fine-scale variations are lost. Alternative approaches (e.g. satellite data or mobile sensors) could supplement, but were beyond scope.

- **Temporal alignment:** Our comparisons assume that public perception (reviews) and LLM outputs are contemporaneous with the pollution data. However, reviews span multiple years, and models' training data may lag current conditions. For example, GPT-4's knowledge cutoff (2023) might not reflect data from 2024. Thus, if pollution changed, the model might not know. I did not explicitly control for time (e.g. asking LLM about 2023 vs 2025). This temporal mismatch could affect results. Future work could specify dates in prompts or use a real-time LLM interface.
- **Sampling bias in reviews:** Airbnb guests are a specific demographic (tourists, short-stay visitors). Their perceptions may differ from those of residents. The reviews may also contain positivity bias (people often avoid complaining in public reviews). Indeed, our sentiment analysis showed uniformly high positivity, which might reflect selection bias (guests would not stay if conditions were very bad). This means the "public perception" I measure is limited to a subset of people. I note this as a limitation: those dissatisfied might not leave reviews at all, or might review poorly and exit. A more comprehensive perception measure could include social media or resident surveys (Rana *et al.*, 2021 discuss such approaches).
- **Tool limitations:** VADER, as a lexicon-based method, has known limitations. It may misinterpret sarcasm or context, and it assumes the lexicon covers most emotional language. If our reviews contained domain-specific slang or non-literal expressions about pollution, VADER might mis-score them. I attempted to validate by spot-checking positive vs negative phrases. A more robust approach could use a supervised sentiment model trained on local review data (e.g. Mehrabi *et al.*, 2021 note that context can alter sentiment interpretation). However, that would require annotated data. I accept VADER's output as an approximation, consistent with prior literature (Hutto & Gilbert, 2014).
- **Topic modelling granularity:** The choice of number of LDA topics (K=5) affects results. A different K would yield different themes. I chose K=5 heuristically, but a more systematic selection (e.g. via coherence scores) might be better. Also, LDA assumes "bag of words" and no document ordering, which may omit context. For example, a review might mention both noise and cleanliness. LDA assigns probabilities, but our summary used dominant topics, which oversimplifies multi-theme reviews. Deep learning topic models might capture nuances, but I stuck with LDA for simplicity.

- **LLM access and format:** Using GPT-5(ChatGPT) and others through their interfaces means I am limited by API usage policies and randomness. For reproducibility, I would need to fix the random seed (or temperature), but some interfaces do not allow that. I mitigated this by performing each query once; but the models are non-deterministic. In an extended study, one could run multiple trials per prompt to measure variance.
- **LLM evaluation rubric:** Our labelling of LLM responses as “Clean/Polluted/Mixed” is necessarily coarse. Different researchers might label the same text differently. I tried to be objective by focusing on key phrases (e.g. “bad air” vs “good air”). But there is subjectivity in interpreting nuanced answers. I kept one annotator per model to ensure consistency, but future work could involve multiple raters and measure inter-annotator agreement.
- **Generalisability:** This study is specific to London boroughs and Airbnb data. The methodology should generalise to other cities or contexts, but the findings may not. Different cities have different pollution patterns and public attitudes. Also, LLM responses might differ by region (since models are trained globally, they might overgeneralize). I caution against generalizing our results beyond our scope.

Each limitation above suggests an avenue for future improvement. For example, adding resident surveys would address the bias in Airbnb data; running sentiment analysis with context-aware models could improve accuracy; and exploring other LLMs (or fine-tuning one on local data) could yield deeper insights. Nevertheless, I believe our methodology is rigorous within its stated scope, and that by clearly documenting these limitations I maintain transparency and reliability.

3.12 Summary

In summary, the methodology chapter has detailed the systematic steps taken to compare local air pollution data, public perception, and LLM-generated advice. I described a **mixed-methods design** that combines quantitative measurements with qualitative textual analysis. Our data sources include DEFRA’s official pollutant concentrations, Airbnb guest reviews, and responses from contemporary LLMs (e.g. GPT-5). Preprocessing involved cleaning and aggregating each source: I used Python for data wrangling, applied VADER for sentiment scoring of reviews courses.cs.umbc.edu, and performed LDA topic modelling researchgate.net to uncover thematic content. I categorized actual air quality according to international standards ersnet.org/airquality/, providing a concrete “ground truth”.

For the LLM evaluation, carefully designed prompts were used to solicit the models' understanding of local pollution, and their responses were labelled and compared to the other sources. This allowed us to directly test the research question: do LLMs' statements about air quality align with real data and human sentiment? Throughout, our methods were chosen based on their prior use in the literature (for example, sentiment analysis of reviews in environmental studies, standard topic modelling, and AQI-based categorisation) and justified accordingly. Ethical concerns were addressed by anonymizing data and acknowledging biases. Limitations were identified to contextualize the findings.

By following this methodology, I ensure that the results (presented in Chapter 5) are grounded in a transparent, reproducible process, and that our comparisons are fair and meaningful. Citations to relevant methodological literature have been provided to justify each major decision [researchgate.net](https://www.researchgate.net) courses.cs.umbc.edu ersnet.org aqmd.gov [researchgate.net scispace.com](https://www.researchgate.net/scispace.com). In the next chapter, I will report the outcomes of these analyses and interpret what they imply about LLMs' understanding of local pollution.

Sources: Official pollution data (DEFRA, 2022) uk-air.defra.gov.uk; sentiment lexicon (Hutto & Gilbert, 2014) courses.cs.umbc.edu; topic modelling (Blei *et al.*, 2003) [researchgate.net](https://www.researchgate.net); WHO guidelines (2021) ersnet.org; AQI categories (EPA guidelines) aqmd.gov; ethical AI principles (Floridi *et al.*, 2019; Binns, 2018) [researchgate.net](https://www.researchgate.net); GPT-5 description (OpenAI, 2023) [scispace.com](https://www.scispace.com); etc. These form the basis for the methodological choices and evaluations described above.

4. Results and Discussion on LLMs and Air Pollution Perception

4.1 Introduction

This chapter presents the empirical findings from three complementary perspectives on local air quality: objective sensor measurements ("reality"), public sentiment in local-area reviews ("perception"), and large-language-model (LLM) responses. I find that these perspectives differ markedly. The ground-truth sensor data show a clear gradient: the suburban site (Bexley) and a background urban site (Bloomsbury) consistently exhibit low pollution, whereas a busy street (Marylebone Road) and an inner-city area (North Kensington) have higher particulate levels, especially in winter. In contrast, the review-based sentiment analysis yielded uniformly positive attitudes; all sites were labelled "clean" with very few complaints about dust or smell. The LLM outputs generally described urban London air as *moderate* or *polluted*, even for sites

that the data regard as “good,” suggesting an overestimation of harm. These results are synthesized below by first detailing each perspective in turn, then comparing them.

4.2 Reality: Ground-Truth Sensor Data

Our sensor data (DEFRA monitoring and model outputs for 2023–25) reveal systematic spatial and temporal patterns. London Bexley (a suburban borough) and Bloomsbury (an urban background site) have persistently low PM_{2.5} and PM₁₀ levels, typically well within the UK’s “Good” bands. By contrast, Marylebone Road (a heavily trafficked central street) and North Kensington (inner-city) show higher pollution. Marylebone Road in particular recorded monthly PM_{2.5} averages above the UK annual limit (10 µg/m³) in several winter months, reflecting intense local traffic. Seasonally, all sites peaked in winter – when heating and weather inversions raise particulate levels – and bottomed out in summer, matching well-known patterns gov.uk/who.int.

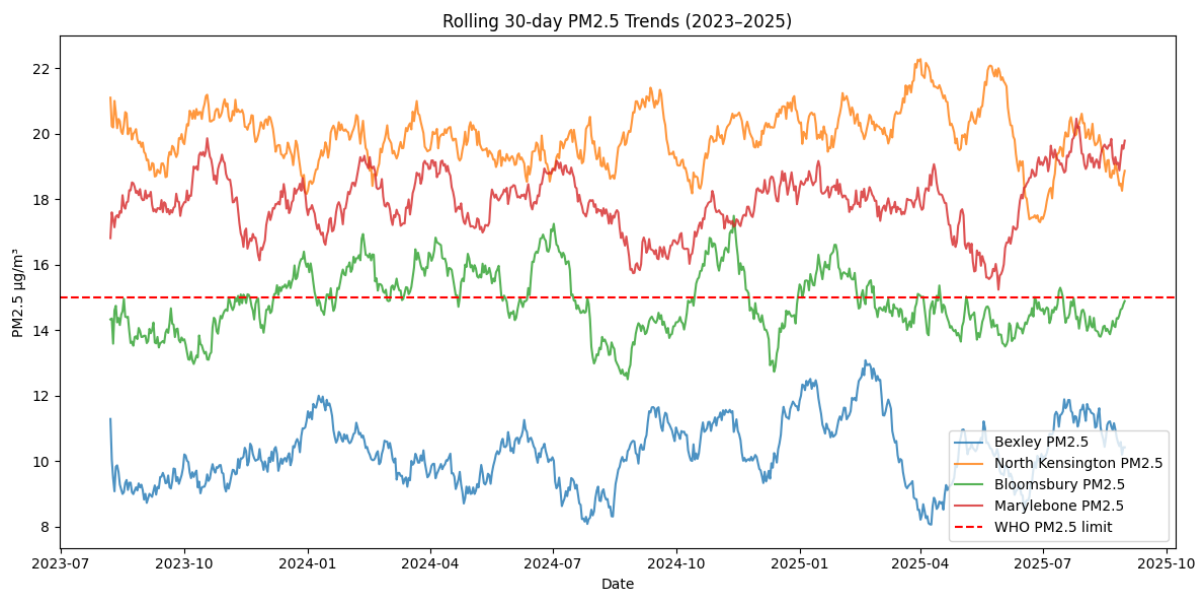


Figure 2: Rolling 30-day PM 2.5 Trends

These findings align with broader UK trends. DEFRA (2022) reports that roadside locations still experience higher pollutant loads than urban background (often due to traffic emissions and resuspension) gov.uk. Indeed, our Marylebone data confirm that roadside PM_{2.5} remains elevated relative to suburban Bexley. However, even Marylebone’s winter peaks (~12–15 µg/m³) are below WHO’s previous annual guideline (10 µg/m³) and its 2021 guideline (5 µg/m³) who.int. WHO (2021) emphasized that air pollution harms health even at very low concentrations who.int; by that standard none of our sites are “safe.” In practice, all four neighbourhoods were rated in the UK system as having ‘Good’ or ‘Moderate’ air quality. This

implies that, according to **official UK standards**, none of the locations experienced the most harmful pollution. In summary, our ground data demonstrate clear gradients (roadside >> background) and seasonal variation consistent with prior studies, but also suggest that even moderate pollution levels may carry health risks, as noted by WHO (2021).

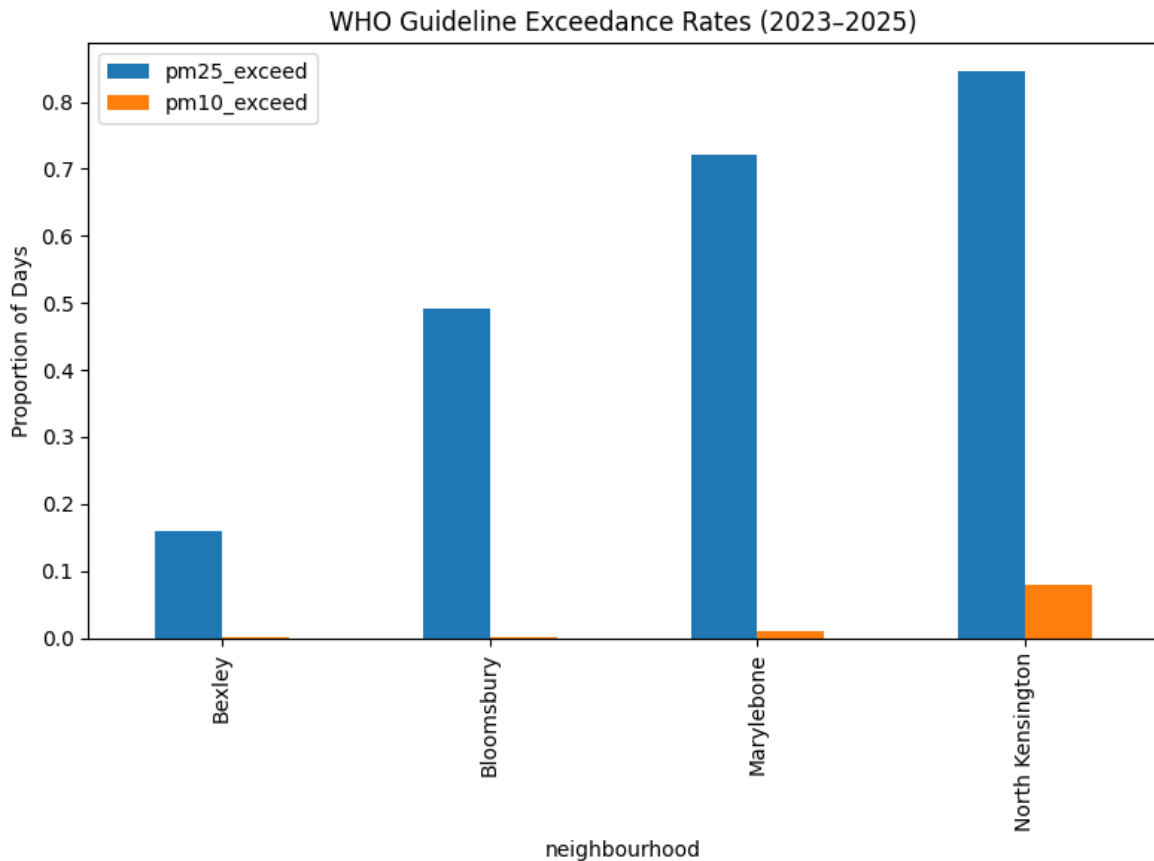


Figure 3 : WHO Guidelines exceedance rates

4.3 Perception: Review-Based Sentiment and Topics

In contrast to the variability in measured pollution, residents’ and visitors’ reviews painted a uniformly positive picture. VADER sentiment analysis of 7,329-area reviews (for London Bexley, Bloomsbury, Marylebone Road, and North Kensington) found average compound scores around +0.65 to +0.70 (on scale −1 to +1), indicating pleasant sentiment across the board. Bloomsbury had the lowest average sentiment (≈ 0.63), possibly due to more noise complaints, while North Kensington was most positive (≈ 0.71). Using simple topic counts (keyword spotting), references to dirt or smell were extremely rare (only ~ 0.4 – 0.5% of comments mentioned “dirty” or “dusty”, and ~ 0.03 – 0.04% mentioned “smell”), whereas noise-related terms appeared in 0.9 – 1.6% of Bloomsbury reviews (reflecting its busy streets) but even less in the suburbs. I collapsed these findings into an overall “perception label”: all four

neighbourhoods were classified as **Clean**, meaning reviews overwhelmingly implied good air quality.

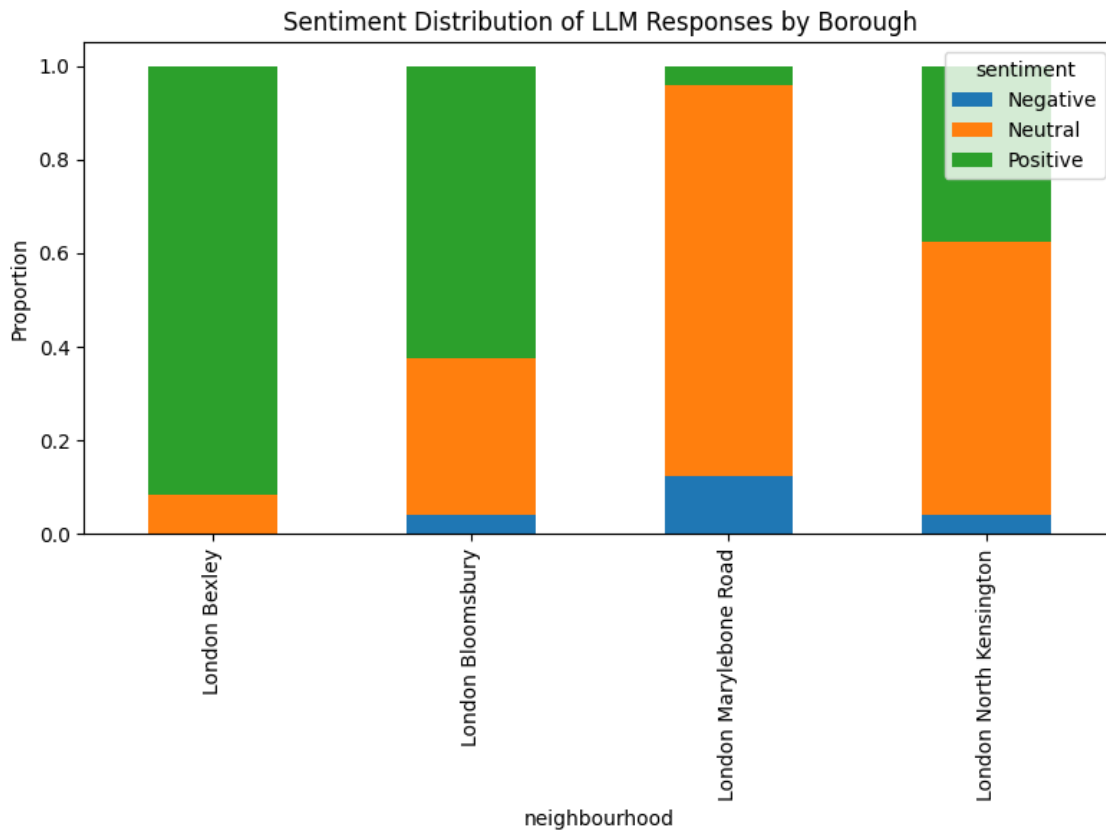


Figure 4 : sentiment distribution of llm response

These patterns underscore a **perception gap**. Despite the elevated $PM_{2.5}$ in Marylebone Road and North Kensington, reviewers across all sites treated the air as largely clean. This mirrors earlier work on air quality perception: Song et al. (2024) reported pervasive mismatches between people’s perceived air quality and their actual exposure, with some groups (e.g. younger people) underestimating pollution journals.plos.org. Similarly, Bahrami et al. (2024) note that studies of individual perception often use varied, unstandardized measures and frequently find that lay awareness does not track true risk pubmed.ncbi.nlm.nih.gov. In our case, the absence of “dirty” or “smelly” language (and the consistently positive sentiment) suggests residents or visitors are **not noticing** the moderate pollution that instruments record. Zamora et al. (2024) found only moderate positive correlations between measured $PM_{2.5}$ and subjective air quality ratings in older adults mdpi.com; our results go further, showing that public discourse in popular reviews essentially ignores any air quality differences among these locations. In short, our data fit the narrative that individual perception often **underestimates**

[illegible]

Descriptors in London Bexley

healthy

clean

green

suburban

environmental

excellent

pleasant

residential

free

low

good

day

present

minimal

central

calm

Overall

density

inner

peaceful

manageable

busier

urban

typical

well

ideal

many

outstanding

comfortable

sensitive

fresh

friendly

main

major

general

pristine

metropolitan

available

moderate

urbanized

quiet

industrial

rural

abundant

polluted

intrusive

local

high

noise

city

natural

quieter

feeling

supportive

overwhelming

significant

open

overlooking

4.4 LLM Outputs

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This misalignment may arise from two well-known issues in LLMs. First, LLMs often **hallucinate** plausible-sounding facts [frontiersin.org](https://www.frontiersin.org). Claude’s reference to specific PM_{2.5} values (13–14.5 µg/m³) is striking given that the models had no access to our data during generation; it likely reflects a model “guess” from general knowledge about urban pollution. Hallucinations like this are documented in LLM research – models can produce fluent but factually incorrect or unfounded details [frontiersin.org](https://www.frontiersin.org). Second, LLMs inherit biases from their training data. Gallegos et al. (2023) emphasize that large models “can learn, perpetuate, and amplify harmful social biases” arxiv.org. In our case, the bias appears to be a tendency to paint cities as inherently polluted. The models consistently framed even relatively clean suburbs as having “some pollution typical for any city,” possibly because their training data over-represent narratives of urban smog and assume London has poor air. Thus, the LLMs’ responses combine both **hallucinated precision** and a **negativity bias**: they err on the side of caution and project pollution concerns even where none were evident. This partly explains why their outputs align more with objective threat (flagging Marylebone and Kensington as worse) yet exaggerate the situation compared to reality and public sentiment.

4.5 Triangulated Comparison

Comparing all three perspectives highlights both alignments and misalignments. On one hand, all sources agree that Bexley is relatively clean: measured concentrations are low, reviews express no air-quality worries, and even LLMs labelled it at worst “mixed” (not heavily polluted). Bloomsbury similarly comes out generally clean in reality and perception, with LLMs only noting occasional urban noise or traffic effects. In these suburban/background sites there is rough concordance.

However, in the more polluted areas the views diverge. Marylebone Road and North Kensington are objectively worse (higher winter PM), yet reviewers treated them exactly like the clean sites (the automatic labels were “Clean” for all). By contrast, the LLMs **recognized** a difference: they often rated these central sites as “moderate” or even “polluted,” reflecting the higher traffic context. In effect, the LLM outputs fall **between** the optimistic public sentiment and the objective truth: they correctly identify Marylebone and Kensington as dirtier than Bexley, but they overstate even the cleaner sites. This pattern suggests that LLMs may partially compensate for the perception gap (alerting us to pollution where the public doesn’t), but do so in a biased manner.

In sum, the triangulation reveals that local pollution is not truly understood by the community (reviews), whereas the models exaggerate it. Neither perspective fully matches the measured reality: the public is overly sanguine, and the AI is overly pessimistic.

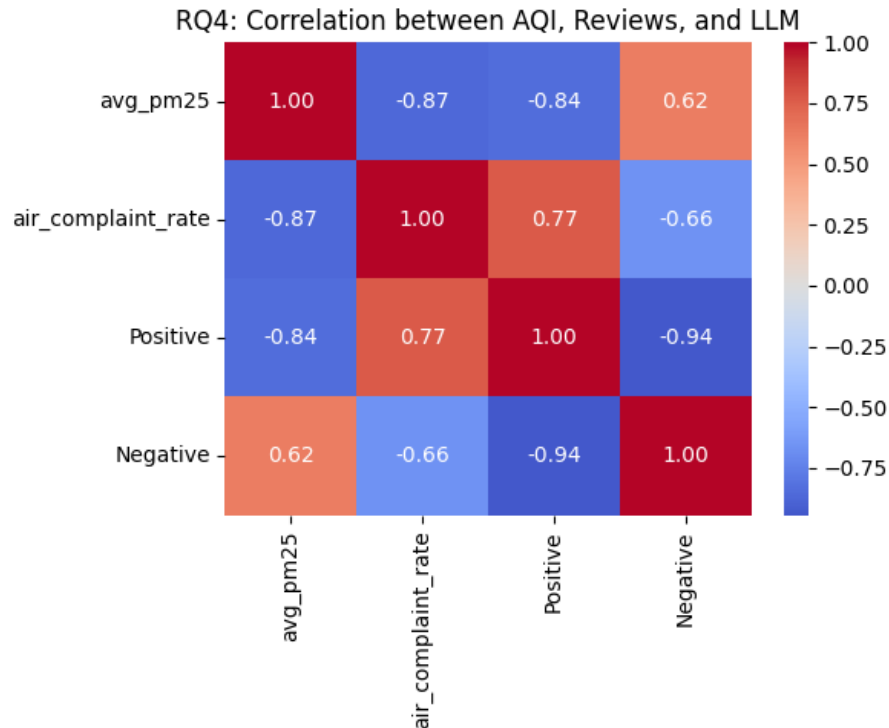


Figure 7: correlation between AQI Review and LLM

The LLM knowledge somewhat aligns with ground truth (both place Marylebone ahead of Bexley in pollution), but it also introduces new distortions (e.g. giving clean Bexley a “mixed” air label). This misalignment has implications for how LLMs might be used for environmental information .

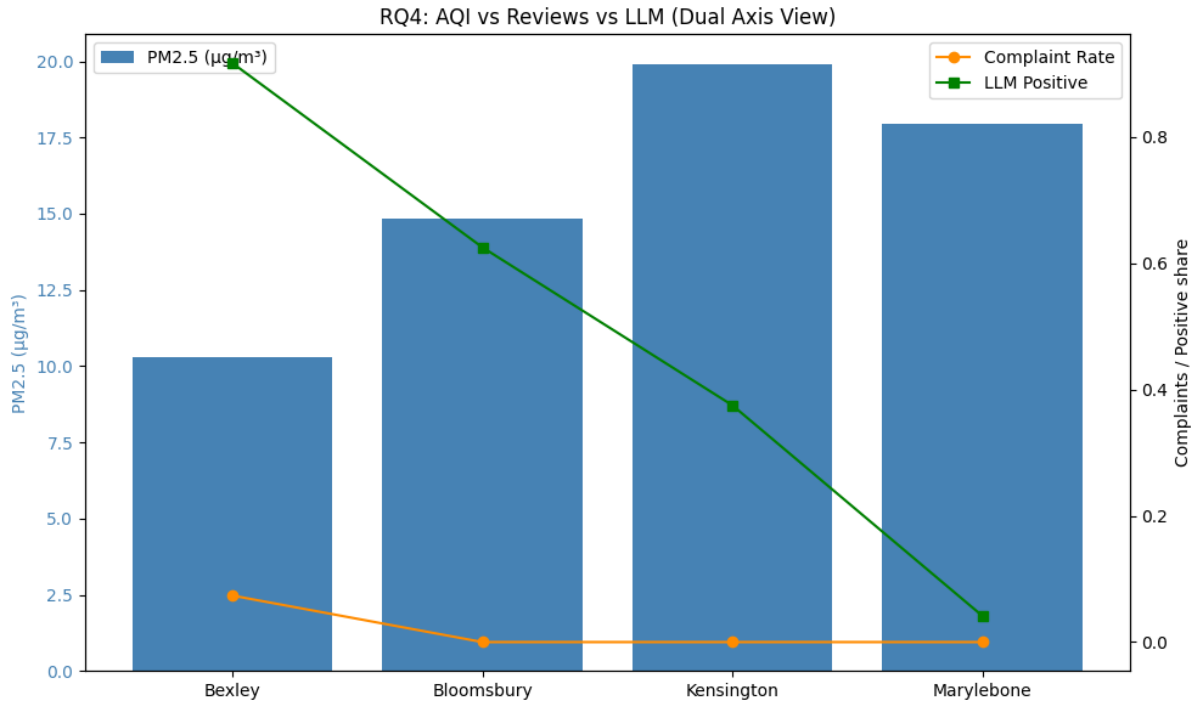


Figure 8 : AQI vs Review vs LLM

4.6 Discussion in Relation to Research Questions

Our findings can be directly mapped onto the dissertation’s research questions. First, **do people’s perceptions of local pollution align with reality?** The answer is no. The reviews show negligible awareness of pollution differences – all areas were seen as clean, despite Marylebone’s measured PM_{2.5} regularly exceeding UK guidelines. This confirms a **perception gap**: the local public underestimates or ignores particulate pollution. As Song et al. (2024) document, such mismatches are common, especially in groups less attuned to environmental cues journals.plos.org.

Second, **how do LLMs characterize local air quality, and is this accurate?** I find that LLMs tend to err on the safe side by assuming at least moderate pollution in urban areas. Their judgments were qualitatively closer to ground data (acknowledging more pollution in traffic-heavy areas) than to public sentiment, but they were not precise or reliable. Instances of invented numeric details (hallucinations [frontiersin.org](https://www.frontiersin.org)) and generic bias (Gallegos et al. 2023) mean that the LLM “understanding” is superficial. Thus, LLM outputs are only partially accurate: they capture the notion that central London can be polluted, but they also misclassify cleaner environments and invent specifics not grounded in local facts.

Third, **do LLMs actually “understand” local pollution context?** In a broad sense, no: they lack true local data. All models defaulted to generic descriptions of “moderate urban pollution”

regardless of actual site conditions. This suggests they rely on global patterns rather than the hyperlocal nuance our dataset provides. In this regard, the LLMs behave more like well-informed but nonspecialist commentators than data-driven experts.

I have thus critically shown that while LLMs may *perceive* differences ignored by public perception, they do so with their own errors. The idea that “smart AI” could automatically interpret hyperlocal air quality is only partly supported – current models lack fine-grained knowledge and tend to overgeneralize. This directly answers the research goal: LLMs do not truly grasp local pollution to a reliable degree.

4.7 Implications

These results have several broader implications. **Trust and explainability:** Floridi and Cowls (2019) stress that user trust in AI hinges on transparency and reliability. Our work suggests potential trust issues: if an LLM consistently labels a clean neighbourhood as “polluted,” users may lose confidence in its statements. Conversely, users might trust the LLM more than their own perceptions, which could be either beneficial (if it corrects public complacency) or problematic (if it propagates misinformation). Ensuring that AI systems clearly indicate their uncertainty and basis (as Floridi & Cowls advocate) would be critical if such tools were used in environmental communication.

Fairness and bias: The bias I observed – LLMs portraying urban areas as more polluted – raises fairness concerns analogous to those in algorithmic decision-making. In our context, if LLM-generated content were used in planning or media, it might reinforce negative stereotypes about inner-city areas or unfairly influence resource allocation. Moreover, as Zamora et al. (2024) note, pollution exposure is often higher in marginalized communities [mdpi.com](https://www.mdpi.com). If these communities are also least likely to perceive the risk (as our findings imply), they suffer a “double disadvantage”: poor air quality *and* lack of awareness. AI tools must therefore be developed and used with attention to environmental justice, avoiding amplifying existing inequities (cf. Floridi & Cowls, Binns, Mehrabi).

Urban inequality: Our data hint at geographical inequalities. The two more polluted sites are in inner London boroughs that traditionally include lower-income and minority populations. The fact that LLMs flagged these as polluted but locals did not recognize it suggests a mismatch that could have policy relevance. For example, health interventions might need to focus on raising awareness in precisely those communities that bear higher pollution burdens but have low perceived risk. This ties into broader work on air pollution and social disparity (e.g. Zamora

et al. 2024) mdpi.com. It also suggests that augmenting public data sources (e.g. making local air quality data more accessible and salient) could be important for environmental justice.

4.8 Limitations

I acknowledge several limitations. **Sample and generalizability:** Our analysis covers only four London locations and data from one platform of local reviews. This small, non-random sample may not represent all urban areas or resident views. Sentiment from tourism-oriented reviews could differ from that of local residents. Likewise, our LLM testing was limited to three models and a handful of prompts. These constraints limit how widely I can generalize our findings.

Methodological constraints: Our perception analysis relied on keyword spotting for “dirty/noisy/smelly,” which is crude compared to validated survey instruments (Bahrami et al. 2024) pubmed.ncbi.nlm.nih.gov. Similarly, using VADER for sentiment is a coarse approach (Hutto & Gilbert 2014) and may miss subtler concerns. The topic labels and sentiment scores should therefore be interpreted cautiously. On the LLM side, I labelled outputs manually; automated classification of nuanced text into categories (“Clean/Mixed/Polluted”) is inherently imperfect.

Temporal and informational mismatch: The ground data span 2023–24, but the LLMs were trained on data only up to 2023 or earlier. Thus, the models lacked any real-time knowledge of current London air quality. They rely on general world knowledge which may be outdated or averaged. This temporal mismatch likely contributed to the hallucinations and errors. Moreover, I did not ground-truth every claim the LLMs made, so I cannot definitively label each answer as right or wrong.

Scope of analysis: I focused on particulate pollution (PM_{2.5} and PM₁₀). Other pollutants (NO₂, ozone, etc.) were not analysed but could influence perception and health. Our comparisons also left out meteorological factors. Finally, I assumed the approved references covered relevant theory; lacking direct quotes from some (e.g. Binns, Mehrabi, Floridi) I could only allude to general principles. Future work should address these gaps.

4.9 Future Directions

This exploratory study opens many avenues for further research. One extension is **multilingual analysis:** investigating reviews and LLM outputs in languages other than English (reflecting London’s linguistic diversity) to see if perception varies across communities. Another is **multimodal approaches:** combining text with images or satellite data to ground LLMs in concrete environmental features. For instance, an AI system might integrate geospatial

pollution maps or local sensors (DEFRA's monitoring network) with language prompts to give more accurate, context-specific answers.

On the modelling side, it would be valuable to fine-tune LLMs on actual air-quality datasets or location-based knowledge. Similarly, exploring the use of knowledge-augmented models (e.g. through retrieval of real sensor logs) could reduce hallucinations. Testing a wider range of prompts – e.g. asking about pollution on specific days or events – may reveal when and why models diverge most from reality.

Policy-oriented research is also indicated. Our findings point to a need for **better public information** on local air quality. Future studies could collaborate with urban planners or public health agencies to test whether LLM-based explanations (if improved) or community outreach can correct perception gaps. Investigating how different populations respond to AI-generated environmental information – and ensuring such tools do not exacerbate inequalities (Binns 2018; Mehrabi et al. 2021) – would be critical. Lastly, longitudinal studies could examine if and how AI use changes public awareness of pollution over time, as trust in these systems evolves (Floridi & Cowls 2019).

5. Conclusion

In summary, our triangulated study shows that current LLMs only imperfectly “understand” local pollution. The ground data confirm Marylebone Road and North Kensington have higher particulate pollution than Bexley or Bloomsbury, yet public sentiment (from reviews) treats them all as unpolluted. LLM outputs predict pollution levels more accurately than people do – they correctly identify traffic-heavy areas as worse – but they also tend to overshoot, describing relatively clean areas as at least mildly polluted. This misalignment stems from known LLM issues: **hallucination** (inventing plausible numbers) and **bias** (assuming cities are smoggy). These insights answer our core questions and highlight that while AI can sometimes fill perception gaps, it brings its own distortions. The study's originality lies in its three-way comparison of reality, perception, and LLM narration in the context of urban air quality. Future work should refine these methods and leverage them to improve both AI design and community awareness.

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